THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Nerve cell is long and branched. Which property is most useful? 1

(a) storing food **(b)** carrying oxygen
(c) carrying signals over distance **(d)** making chlorophyll

2. Which organelle is absent in an amoeba but present in a leaf cell? 1

(a) nucleus **(b)** cell membrane **(c)** chloroplast **(d)** cytoplasm

3. Which sequence correctly shows increasing size? 1

(a) amoeba, virus, bacterium, egg **(b)** virus, bacterium, amoeba, human egg
(c) human egg, bacterium, virus, amoeba **(d)** bacterium, virus, egg, amoeba

4. Which of the following will you observe if you view a drop of pond water under a microscope? 1

(a) only crystals **(b)** moving organisms like Paramecium
(c) oxygen gas **(d)** nothing at all

5. Fermentation of sugar by yeast produces: 1

(a) carbon dioxide and alcohol **(b)** only water
(c) oxygen and water **(d)** only oxygen

6. An animal cell is placed in water and swells but does not burst in a plant cell because of the: 1

(a) rigid cell wall **(b)** nucleus **(c)** vacuole **(d)** cytoplasm

7. Cells of a mushroom lack: 1

(a) nucleus **(b)** cell membrane **(c)** cytoplasm **(d)** chlorophyll

8. Which of these cannot be treated with antibiotics? 1

(a) influenza (viral) **(b)** cholera
(c) tuberculosis **(d)** typhoid

9. Which of the following does NOT occur in a unicellular organism? 1

(a) different organs for each function **(b)** reproduction
(c) excretion **(d)** digestion

10. A student marks 'plasma membrane' in a drawing of an onion cell. The membrane lies: 1

(a) outside the wall **(b)** just inside the cell wall
(c) inside the nucleus **(d)** at the centre

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can a single-celled organism like Amoeba live without organs? 2

12. Why is a red blood cell without a nucleus advantageous? 2

13. Why do leaf cells contain more chloroplasts than root cells? 2

14. How is yeast used to make bread soft and spongy? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the structure of a root hair cell suits its function. 3

16. What are decomposers? Name a bacterium and a fungus that act as decomposers. 3

17. A bacterium and an amoeba are both single-celled. Give three differences. 3

OR

What do you understand by specialisation of cells? Give two examples. 3

18. Why does a mixture of flour, sugar and yeast show frothing when kept in a warm place? 3

OR

What is meant by 'host' and 'parasite'? Give an example of a virus-host pair. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the lifestyle and structure of yeast. How does it help in baking and brewing? 5

OR

Explain three differences between the plant and animal cell. Why does a plant cell have a regular shape? 5

20. Explain the specialisation of cells with five examples of cells from plants and animals. 5

OR

Explain four ways in which microorganisms are harmful, with one example each. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND